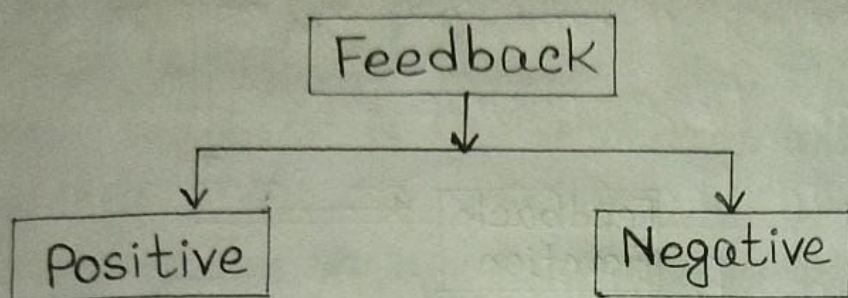


Amplifiers with Negative Feedback

▣ What is feedback?

Ans: The process of injecting a fraction of output energy of some device back to the input is known as feedback.



▣ What is positive feedback?

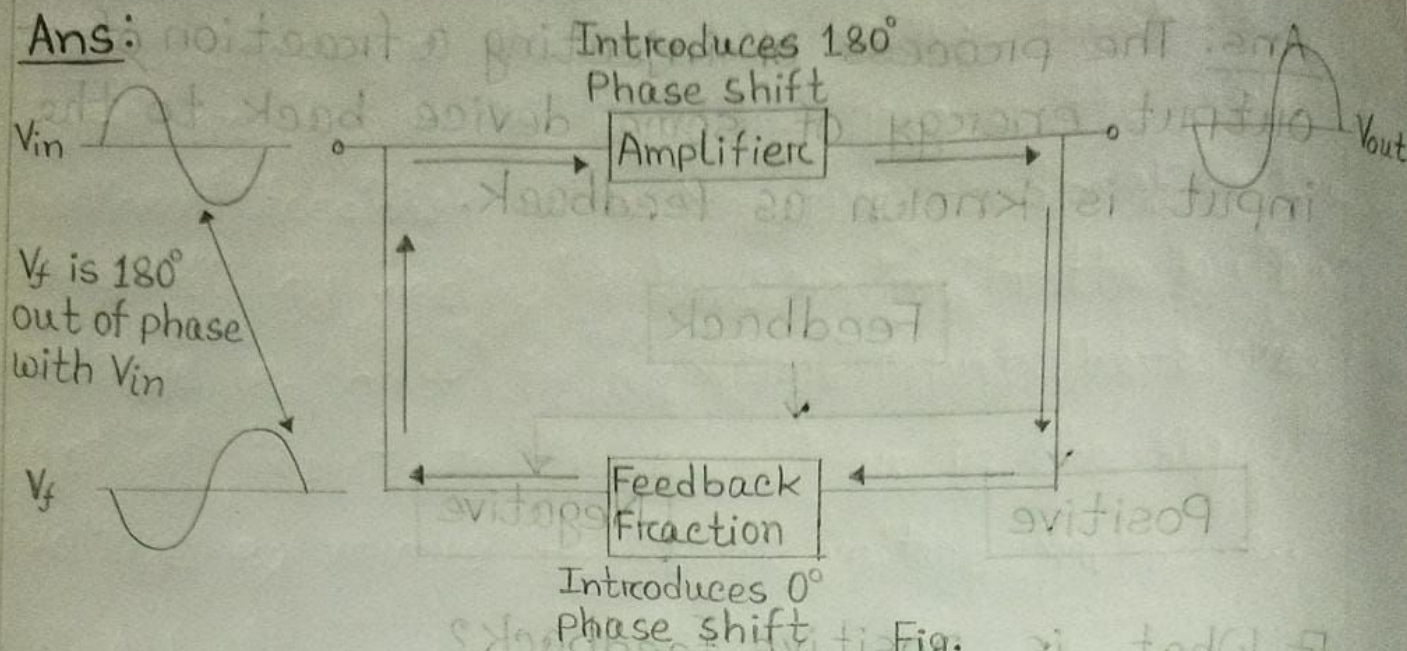
Ans: When the feedback energy (voltage or current) is in phase with the input signal and thus aids it, it is called positive feedback.

▣ What is negative feedback?

Ans: When the feedback energy (voltage or current) is out of phase with the input signal and thus opposes it, it is called negative feedback.

Derive the principle of negative feedback in amplifier.

Ans:



A feedback amplifier has two parts viz an amplifier and a feedback circuit. The feedback circuit usually consists of resistors and returns a fraction of output energy back to the input. Fig. shows the principles of negative voltage feedback in an amplifier. Typical values have been assumed to make the treatment more illustrative. The output of the amplifier is 10V. The fraction m_v of this output i.e. 100 mV is feedback to the input where it is applied in series with the input signal of 101 mV. As the feedback is negative, therefore, only 1 mV appears at the input terminals of the amplifier. Referring to Fig. , we have, gain of amplifier without feedback,

$$A_v = \frac{10\text{ V}}{1\text{ mV}} = 10,000$$

Fraction of output voltage feedback, $m_v = \frac{100\text{ mV}}{10\text{ V}} = 0.01$

Gain of amplifier with negative feedback, $A_{vf} = \frac{10\text{ V}}{101\text{ mV}} = 100$

The following points are worth noting:

- i. When negative voltage feedback is applied, the gain of the amplifier is reduced. Thus, the gain of above amplifier without feedback is 10,000 whereas with negative feedback, it is only 100.
- ii. When negative voltage feedback is employed, the voltage actually applied to the amplifier is extremely small. In this case, the signal voltage is 101 mV and the negative feedback is 100 mV so that voltage applied at the input of the amplifier is only 1 mV.
- iii. In a negative voltage feedback circuit, the feedback fraction m_v is always between 0 and 1.
- iv. The gain with feedback is sometimes called closed-loop gain while the gain without feedback is called open-loop gain. These terms come from the fact that amplifier and feedback circuits form a "loop". When the loop is "opened" by disconnecting the feedback circuit from the input, the amplifier's gain is A_v , the "open-loop" gain. When the loop is "closed" by connecting the feedback circuit, the gain decreases to A_{vf} , the "closed-loop" gain.

Q Derive the gain of negative voltage feedback amplifier.

Ans: Consider the negative voltage feedback amplifier shown in Fig.1. The gain of the amplifier without feedback is A_v . Negative feedback is then applied by feeding a fraction m_v of the output voltage e_o back to amplifier input. Therefore, the actual input to the amplifier is the signal voltage e_g minus feedback voltage $m_v e_o$ i.e.,

$$\text{Actual input to amplifier} = e_g - m_v e_o$$

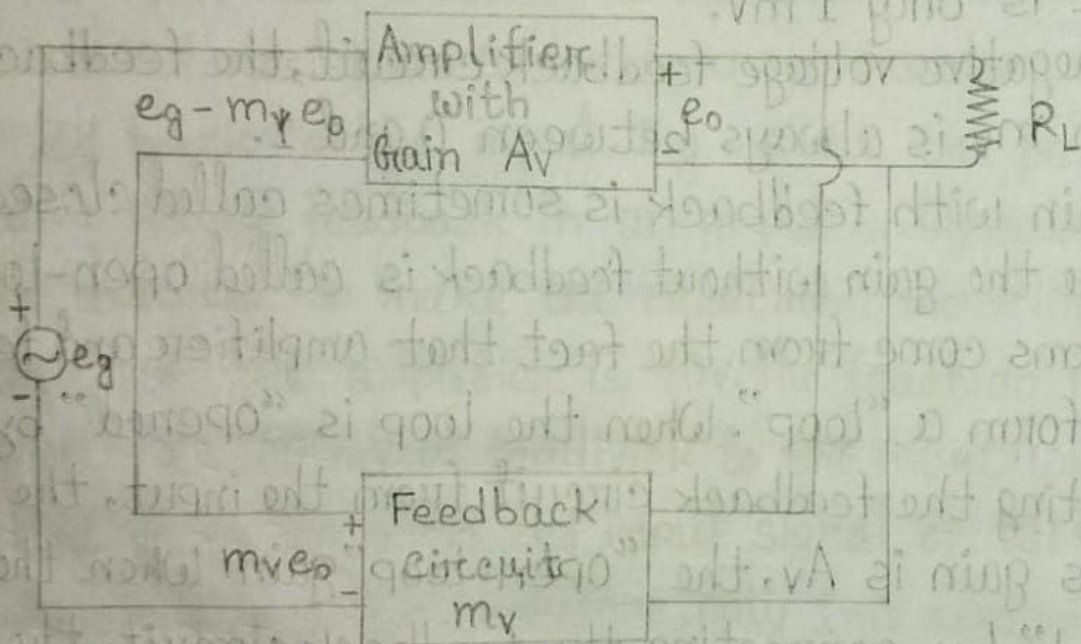


Fig. 1

The output e_o must be equal to the input voltage $e_g - m_v e_o$ multiplied by gain A_v of the amplifier i.e.,

$$(e_g - m_v e_o) A_v = e_o$$

$$\text{or, } A_v e_g - A_v m_v e_o = e_o$$

$$\text{or, } e_o + A_v m_v e_o = A_v e_g$$

$$\text{or, } e_o (1 + A_v m_v) = A_v e_g$$

$$\text{or, } \frac{e_o}{e_g} = \frac{A_v}{1 + A_v m_v}$$

But $\frac{e_o}{e_g}$ is the voltage gain of the amplifier with feedback.

$\therefore$ Voltage gain with negative feedback is,

$$A_{vf} = \frac{A_v}{1 + A_v m_v}$$

Q. Show that, the input impedance of an amplifier increases due to negative voltage feedback.

Ans:

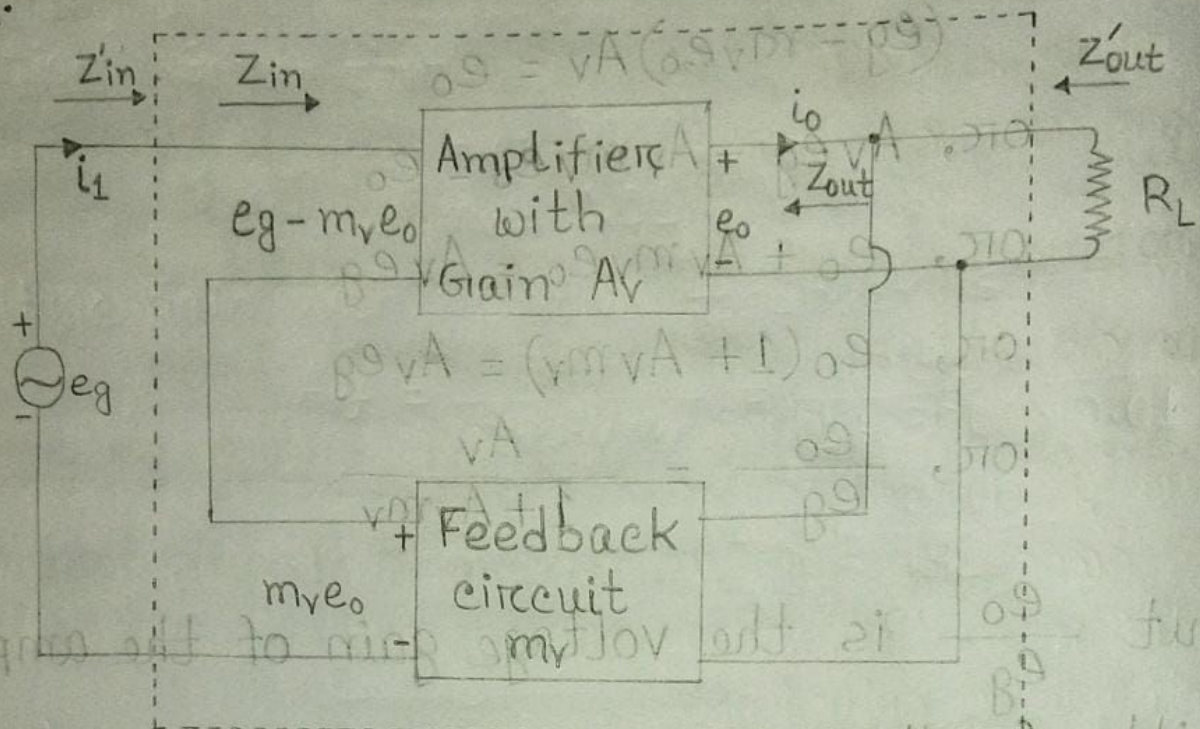


Fig. 2

The increase in input impedance with negative voltage feedback can be explained by referring to Fig. 2. Suppose the input impedance of the amplifier is Z_{in} without feedback and Z'_in with negative feedback. Let us further assume that input current is i_1 .

Referring to Fig., we have,

$$e_g - m_v e_o = i_1 Z_{in}$$

Now,

$$e_g = (e_g - m_v e_o) + m_v e_o$$

$$\text{or, } e_g = (e_g - m_v e_o) + A_v m_v (e_g - m_v e_o)$$

$$[\because e_o = A_v (e_g - m_v e_o)]$$

$$\text{or, } e_g = (e_g - m_v e_o) (1 + A_v m_v)$$

$$\text{or, } e_g = i_1 Z_{in} (1 + A_v m_v)$$

$$[\because e_g - m_v e_o = i_1 Z_{in}]$$

$$\text{or, } \frac{e_g}{i_1} = Z_{in} (1 + A_v m_v)$$

But $\frac{e_g}{i_1} = Z'_{in}$, the input impedance of the amplifier with negative voltage feedback.

$$\therefore Z'_{in} = Z_{in} (1 + A_v m_v).$$

[showed]

☐ Why negative feedback used in amplifiers?

Ans: Negative feedback reduces the gain of the amplifiers. It also reduces distortion, noise and instability. It increases bandwidth and improves input and output impedances. Due to these advantages, the negative feedback is frequently used in amplifiers.

Example: The voltage gain of an amplifier without feedback is 3000. Calculate the voltage gain of the amplifier if negative voltage feedback is introduced in the circuit. Given that feedback fraction $m_v = 0.01$

Solution: Given that,

$$A_v = 3000$$

$$m_v = 0.01$$

$\therefore$ Voltage gain with negative feedback is

$$\begin{aligned} A_{vf} &= \frac{A_v}{1 + A_v m_v} = \frac{3000}{1 + 3000 \times 0.01} \\ &= \frac{3000}{31} \\ &= 97 \end{aligned}$$

Example: The overall gain of a multistage amplifier is 140. When negative voltage feedback is applied, the gain is reduced to 17.5. Find the fraction of the output that is feedback to the input.

Solution: Given that,

$$A_v = 140$$

$$A_{vf} = 17.5$$

Let m_v be the feedback fraction. Voltage gain with negative feedback is

$$A_{vf} = \frac{A_v}{1 + A_v m_v}$$

$$\text{or, } 17.5 = \frac{140}{1 + 140 m_v}$$

$$\text{or, } 17.5 + 2450 m_v = 140$$

$$\text{or, } m_v = \frac{140 - 17.5}{2450}$$

$$\text{or, } m_v = 0.05$$

$$\therefore m_v = \frac{1}{20}$$

Example: When negative voltage feedback is applied to an amplifier of gain 100, the overall gain falls to 50.

(i) calculate the fraction of the output voltage feedback.

(ii) If this fraction is maintained, calculate the value of the amplifier gain required if the overall stage gain is to be 75.

Solution: (i) Given that,

Gain without feedback, $A_v = 100$

Gain with feedback, $A_{vf} = 50$

Let m_v be the fraction of the output voltage feedback.

$$\text{Now, } A_{vf} = \frac{A_v}{1 + A_v m_v}$$

$$\text{or, } 50 = \frac{100}{1 + 100 m_v}$$

$$\text{or, } 50 + 5000 m_v = 100$$

$$\text{or, } m_v = \frac{100 - 50}{5000}$$

$$\text{or, } m_v = 0.01$$

(ii) Given that,

$$A_{vf} = 75$$

$$m_v = 0.01$$

$$\text{Now, } A_{vf} = \frac{A_v}{1 + A_v m_v}$$

$$\text{or, } 75 = \frac{A_v}{1 + 0.01 A_v}$$

$$\text{or, } 75 + 0.75 A_v = A_v$$

$$\text{or, } A_v - 0.75 A_v = 75$$

$$\text{or, } A_v = \frac{75}{0.25}$$

$$\therefore A_v = 300.$$

Example: With a negative voltage feedback, an amplifier gives an output of 10V with an input of 0.5 V. When feedback is removed, it requires 0.25 V input for the same output. Calculate (i) gain without feedback (ii) feedback fraction m_v .

Solution: (i) Gain without feedback, $A_v = \frac{10}{0.25} = 40$

(ii) Gain with feedback, $A_{vf} = \frac{10}{0.5} = 20$

$$\text{Now, } A_{vf} = \frac{A_v}{1 + A_v m_v}$$

$$\text{or, } 20 = \frac{40}{1 + 40m_v}$$

$$\text{or, } 20 + 800m_v = 40$$

$$\text{or, } m_v = \frac{40 - 20}{800}$$

$$\text{or, } m_v = 0.025$$

$$\therefore m_v = \frac{1}{40}$$

Example: The gain of an amplifier without feedback is 50 whereas with negative voltage feedback, it falls to 25. If due to ageing, the amplifier gain falls to 40, find the percentage reduction in stage gain (i) without feedback and (ii) with negative feedback.

Solution: Given that,

Gain without feedback, $A_v = 50$

Gain with negative feedback, $A_{vf} = 25$

$$\text{Now, } A_{vf} = \frac{A_v}{1 + A_v m_v}$$

$$\text{or, } 25 = \frac{50}{1 + 50 m_v}$$

$$\text{or, } 25 + 1250 m_v = 50$$

$$\therefore m_v = \frac{1}{50}$$

(i) Without feedback: The gain of the amplifier without feedback is 50. However, due to ageing, it falls to 40.

$$\therefore \% \text{ age reduction in stage gain} = \frac{50 - 40}{50} \times 100 = 20\%$$

(ii) With negative feedback: When the gain without feedback was 50, the gain with negative feedback was 25. Now the gain without feedback falls to 40.

$$\therefore \text{New gain with negative feedback} = \frac{A_v}{1 + A_v m_v} = \frac{40}{1 + (40 \times \frac{1}{50})} = 22.22$$

$$\therefore \% \text{ age reduction in stage gain} = \frac{25 - 22.2}{25} \times 100 = 11.12\%$$